Example Task 1 Aspect Based Sentiment analysis

- This project is based on subtask 2 of SemEval-2014 Task 4: Aspect Based Sentiment Analysis. http://alt.qcri.org/semeval2014/task4/
- You need to design methods for sentiment classification specific to an aspect.
- For example
  - "Even though its good **seafood**, the **prices** are too high".
  - This sentence contains two aspects, namely "**seafood**" and "**prices**". The sentiment for the two aspects are positive and negative respectively.

- **Task:** Predict the aspect-based sentiment polarity for each instance in the test files, and write your predicted labels in result files. The labels in test files are only for evaluation
- **Datasets:** https://alt.qcri.org/semeval2014/task4/index.php?id=data-and-tools Each sample contains
  - id: the sentence id
  - text: the text content
  - aspect:start-end: "aspect" denotes the aspect term mentioned in the text, and "start-end" denotes the start and end character index of the aspect term occurred in the text
  - label: the sentiment label (positive, negative, and neutral) specific to an aspect

Example Task 2 Fake News Challenge

- This project is borrowed from a public challenge for fake news detection. Details can be found in http://www.fakenewschallenge.org/ including the dataset.
  - Report and implementation of the top-3 teams can be found at the bottom.
  - You are required to implement at least one of their models. A novel model for this task is highly encouraged.

FORMAL DEFINITION

Input

A headline and a body text - either from the same news article or from two different articles.

Output

Classify the stance of the body text relative to the claim made in the headline into one of four categories:

1. **Agrees**: The body text agrees with the headline.
2. **Disagrees**: The body text disagrees with the headline.
3. **Discusses**: The body text discuss the same topic as the headline, but does not take a position
4. **Unrelated**: The body text discusses a different topic than the headline

FNC-I RESULTS

Fifty of the 80 participants made submissions for FNC-1 by the end of June 2nd using a wide array of techniques. The teams got access to the (unlabeled) test data and were scored automatically using the Codalab submission platform. The scoring system produces a raw score based on the differentially weighted scoring metric. The relative score the raw score normalized by the maximum possible score on the test set.

Rank	Team name	Affliation	Score	Relative Score
1	SOLAT in the SWEN	Talos Intelligence	9556.50	82.02
2	Athene (UKP Lab)	TU Darmstadt	9550.75	81.97
3	UCL Machine Reading	UCL	9521.50	81.72

Congratulations to our top-3 teams! The top-3 teams will also get a cash prize of USD 1000, USD 600, and USD 400 respectively. In addition to the top-3 teams, we would also like to give a special shoutout to ranks 4 and 5 held by teams at UIUC and U. Arizona. For a complete leaderboard, visit the competition's Codalab page.

Example Task 3 SemEval 2022 Task 8: Multilingual News Article Similarity

- The task is: **Given a pair of news articles, are they covering the same news story**?
- This SemEval task aims to develop systems that identify multilingual news articles that provide similar information
- This is a document-level similarity task in the applied domain of news articles, rating them pairwise on a 4-point scale from most to least similar
- https://competitions.codalab.org/competitions/33835
- Paper: https://aclanthology.org/2022.semeval-1.155.pdf
- Example code: https://github.com/upura/semeval2022-task8-multilingual-news-article-similarity

SemEval-2022 Task 8: Multilingual news article similarity

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This year: https://semeval.github.io/SemEval2024/

Competition

SemEval 2022 Task 8: Multilingual News Article Similarity

Organized by davidjurgens - Current server time: April 2, 2024, 12:45 p.m. UTC

Previous	▶ Current	End
Evaluation	Post-Evaluation	Competition Ends
Jan. 10, 2022, midnight UTC	Feb. 3, 2022, midnight UTC	Never

Learn the Details | Phases | Participate | Results

- Overview
- Evaluation
- The Data
- Schedule
- News pair annotation examples
- Organizers
- Terms and Conditions
- Task Results

Trial Data

Trial data is available here.

Scores are computed by averaging several annotators' scores or by taking one annotator's score if only one is available/valid. In the former case, floats for the 4-point scale are to be expected while in the latter, only integers are possible. Both cases will appear in the training and evaluation data.

News articles are released as URLs to the publisher's website. To facilitate re-hydrating the textual content of the articles, we offer a python script available here. The script attempts to download the earliest available version of the article from the Internet Archive. For articles that are unavailable from the Internet Archive, the script uses newspaper3k to attempt downloading them from the original site of publication.

Training Data

Training data will be released in batches throughout the preparation period. Scores are computed by averaging several annotators' scores. Pairs have between 1 and 8 annotators with the bulk of items having 1-3 annotations. We do not anticipate that scores for previously released items will change with future updates, though we reserve the possibility of updating (e.g., to correct errors). Data will include pairs in the same language as well as pairs in different languages. Two new column fields have been added to the format to reflect the languages of each url (url1_lang, url2_lang).

- **2021/09/15**: Training data v0.1: This release consists of 2,939 pairs of news articles from the following languages (with counts): en: 1761, es: 444, de: 414, pl: 186, tr: 54, ar: 52, and fr: 28. All pairs have the same language in this release.
- **2021/11/04**: Training data v0.2: This release consists of 4,964 pairs of news articles including those in the v0.1 release. This release features new cross-lingual pairs and additional monolingual data. The following are counts of language-pairs released in this data: en-en: 1800, de-de: 857, de-en: 577, es-es: 570, tr-tr: 465, pl-pl: 349, ar-ar: 274, fr-fr: 72

Example Task 4 Commonsense Validation and Explanation (Task A)

- This project is based on the ACL 2019 paper

  Does It Make Sense? And why? A Pilot Study for Sense Making and Explanation

- There are two subtasks: task A and task B. You can choose either one for your project or both.
- You are required to implement at least one of the models except Random in the paper, or devise your own methods to improve the performance.
- **Datasets**: https://github.com/wangcunxiang/SemEval2020-Task4-Commonsense-Validation-and-Explanation

- **Task A**: Commonsense Validation
  - Choose from two natural language statements to judge which one is against commonsense.
- **Example**:
  - Which statement of the two is against commonsense?
  - Statement 1: He put a turkey into the fridge.
  - Statement 2: He put an elephant into the fridge (against commonsense).
- **Input**
  - Two statements (sent0, sent1) with similar words.
- **Output**
  - 0 or 1
  - 0 means sent 1 is against commonsense, while 1 means sent 2 is against commonsense.

Example Task 5 Commonsense Validation and Explanation (Task B)

- This project is based on the ACL 2019 paper

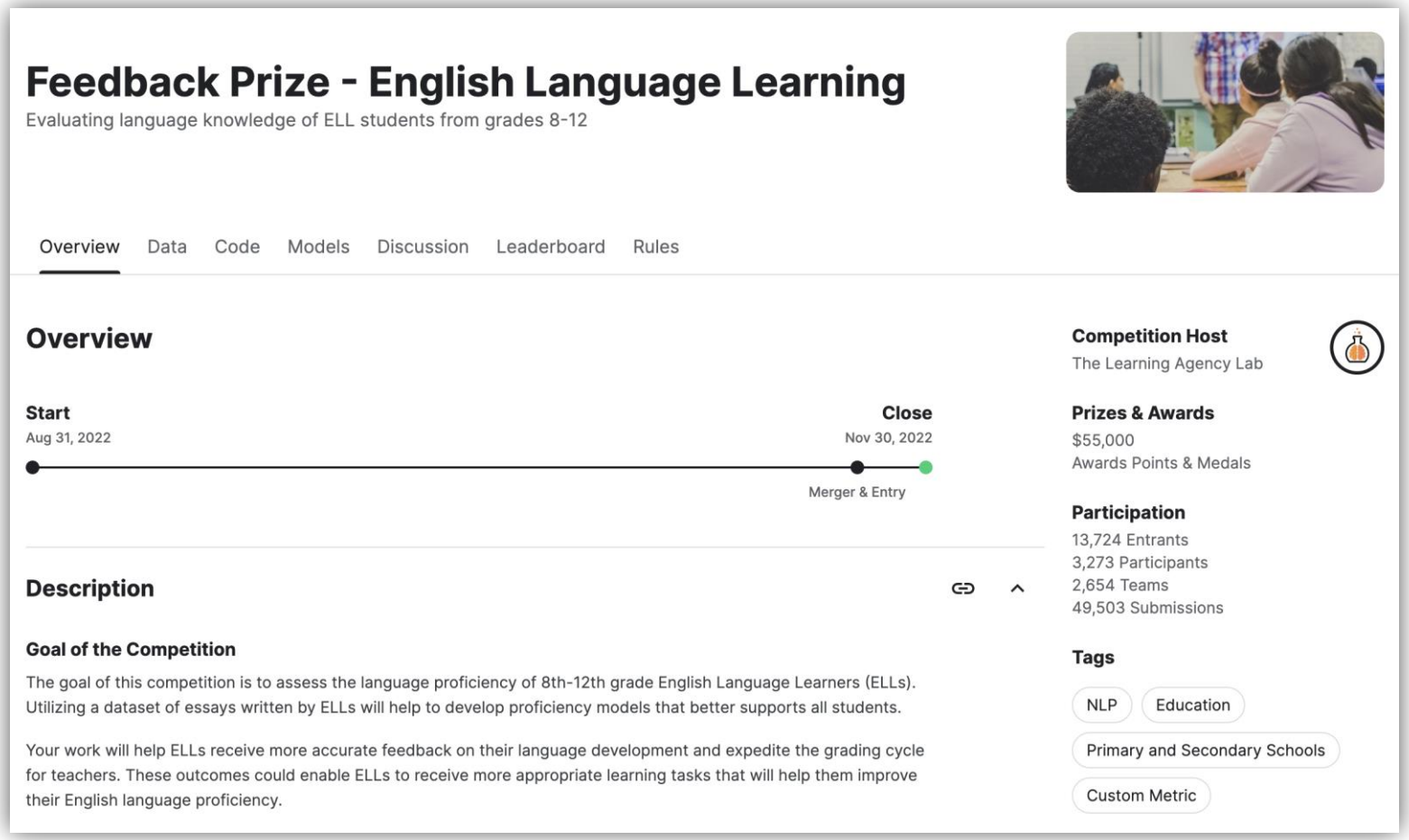
Does It Make Sense? And why? A Pilot Study for Sense Making and Explanation

- There are two subtasks: task A and task B. You can choose either one for your project or both.
- You are required to implement at least one of the models except Random in the paper, or devise your own methods to improve the performance.
- **Datasets**: https://github.com/wangcunxiang/SemEval2020-Task4-Commonsense-Validation-and-Explanation

- **Task B**: Commonsense Explanation
  - Find the key reason from three options to explain why a given statement does not make sense.
- **Example**
  - Statement: He put an elephant into the fridge.
  - Reasons
    - A: An elephant is much bigger than a fridge. (correct explanation)
    - B: Elephants are usually white while fridges are usually white.
    - C: An elephant cannot eat a fridge.
- **Input**
  - One statement against commonsense and three options for explanation.
- **Output**
  - A, B or C

Example Task 6 Feedback Prize - English Language Learning

- The task is: **Assess the language proficiency of 8th-12th grade English Language Learners (ELLs)**
- This task aims to receive more accurate feedback on their language development and expedite the grading cycle for teachers.
- https://www.kaggle.com/competitions/feedback-prize-english-language-learning/data
- Example solution: https://www.kaggle.com/competitions/feedback-prize-english-language-learning/code



Feedback Prize - English Language Learning

Evaluating language knowledge of ELL students from grades 8-12

Overview Data Code Models Discussion Leaderboard Rules

Overview

Start
Aug 31, 2022

Close
Nov 30, 2022

Merger & Entry

Competition Host
The Learning Agency Lab

Prizes & Awards
$55,000
Awards Points & Medals

Participation
13,724 Entrants
3,273 Participants
2,654 Teams
49,503 Submissions

Tags
NLP
Education
Primary and Secondary Schools
Custom Metric

Description

Goal of the Competition

The goal of this competition is to assess the language proficiency of 8th-12th grade English Language Learners (ELLs). Utilizing a dataset of essays written by ELLs will help to develop proficiency models that better supports all students.

Your work will help ELLs receive more accurate feedback on their language development and expedite the grading cycle for teachers. These outcomes could enable ELLs to receive more appropriate learning tasks that will help them improve their English language proficiency.

Example Task 7 CCL 2023 电信网络诈骗案件分类评测

- **本任**务目标：对给定案件描述文本进行分类。**案件文本包含**对案件的整体描述（经过脱敏处理）。**从反**诈大数据平台共计导出 13 **个**类别**的数据，去除了"其他**类型诈骗"类别，因此最终采用 12 **个**类别。

- **目的在于**统计现状，有助于公安部门掌握当前电信网络诈骗案件的分布特点，进而能够对不**同**类别的诈骗案件作出针对性的预防、监管、**制止**、侦查等措施。

- https://github.com/GJSeason/CCL2023-FCC

- Paper: https://aclanthology.org/2023.ccl-3.21.pdf

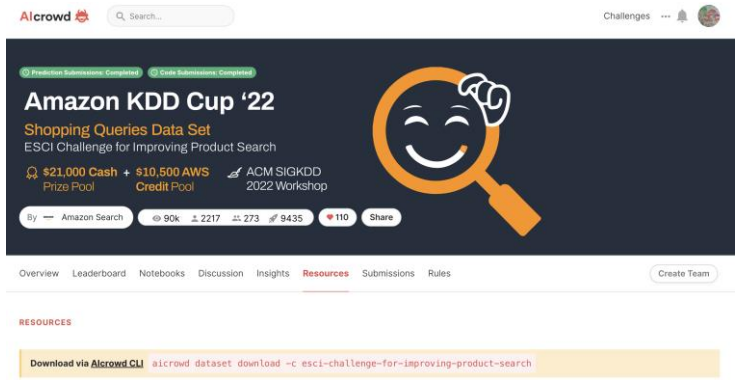
类别名称	样本数量
刷单返利类	35459
冒充电商物流客服类	13772
虚假网络投资理财类	11836
贷款、代办信用卡类	11105
虚假征信类	8464
虚假购物、服务类	7058
冒充公检法及政府机关类	4563
冒充领导、熟人类	4407
网络游戏产品虚假交易类	2155
网络婚恋、交友类（非虚假网络投资理财类）	1654
网黑案件	1197
冒充军警购物类	1092
总计	102762


```
{
    "案件编号": 49750,
    "案情描述": "2022 年 11 月 13 日 14 时 10 分 23 秒我滨河派出所接到 110 报警称在接到自称疾控中心诈骗电
    "案件类别": "冒充公检法及政府机关类"
},
{
    "案件编号": 78494,
    "案情描述": "2022 年 1 月 10 日 11 时至 18 时许，受害人在的家中，接到陌生电话：（对方号码:）对方自称是银
    "案件类别": "虚假征信类"
}
```

Example Task 8 Multiclass Product Classification

- **Task**: Given a query and a result list of products retrieved for this query, the goal of this task is to classify each product as being an Exact, Substitute, Complement, or Irrelevant match for the query.

- https://www.aicrowd.com/challenges/esci-challenge-for-improving-product-search



Shopping Queries Dataset: A Large-Scale ESCI Benchmark for Improving Product Search

Chandan K. Reddy, Lluís Màrquez, Fran Valero, Nikhil Rao, Hugo Zaragoza, Sambaran Bandyopadhyay, Arnab Biswas, Anlu Xing, Karthik Subbian
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{ckreddy,lluismv,fvalero,nikhilsr,hugzarag,sambarab,abisway,anluxing,ksubbian}@amazon.com

ABSTRACT

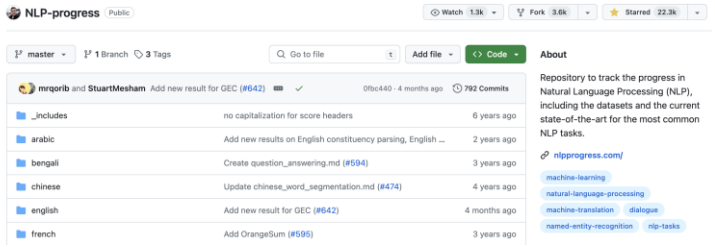
Improving the quality of search results can significantly enhance users experience and engagement with search engines. In spite of several recent advancements in the fields of machine learning and data mining, correctly classifying items for a particular user search query has been a long standing challenge, which still has a large room for improvement. This paper introduces the "Shopping Queries Dataset", a large dataset of difficult Amazon search queries and results, publicly released with the aim of fostering research in improving the quality of search results. The dataset contains around 130 thousand unique queries and 2.6 million manually labeled (query,product) relevance judgements. The dataset is multilingual with queries in English, Japanese, and Spanish. The Shopping Queries Dataset is being used in one of the KDDCup'22 challenges. In this paper, we describe the dataset and present three evaluation tasks along with baseline results: (i) ranking the results list, (ii) classifying product results into relevance categories, and (iii) identifying substitute products for a given query. We anticipate that this data will become the gold standard for future research in

"iPhone", would an iPhone charger be relevant, irrelevant, or somewhere in between? In fact, many users issue the query "iPhone" to find and purchase a charger for the iPhone. They simply expect the search engine to understand their need. For this reason, we break down relevance into the following four classes which are used to measure the relevance of items in the search results:

- **Exact (E):** the item is relevant for the query, and satisfies all the query specifications (e.g., a water bottle matching all attributes of a query "plastic water bottle 24oz", such as material and size)
- **Substitute (S):** the item is somewhat relevant, i.e., it fails to fulfill some aspects of the query but the item can be used as a functional substitute (e.g., fleece for a "sweater" query)
- **Complement (C):** the item does not fulfill the query, but could be used in combination with an exact item (e.g., track pants for "running shoes" query)
- **Irrelevant (I):** the item is irrelevant, or it fails to fulfill a central aspect of the query (e.g., socks for a "telescope" query, or a wheat flour bread for a "gluten–free bread" query)

Example Task 9 Other Tasks (Ruder's list)

- https://github.com/sebastianruder/NLP-progress



Word Sense Disambiguation

The task of Word Sense Disambiguation (WSD) consists of associating words in context with their most suitable entry in a pre-defined sense inventory. The de-facto sense inventory for English in WSD is WordNet. For example, given the word "mouse" and the following sentence:

"A mouse consists of an object held in one's hand, with one or more buttons."

we would assign "mouse" with its electronic device sense (the 4th sense in the WordNet sense inventory).

Fine-grained WSD:

The Evaluation framework of Raganato et al. 2017 [1] includes two training sets (SemCor-Miller et al., 1993- and OMSTI-Taghipour and Ng, 2015-) and five test sets from the Senseval/SemEval series (Edmonds and Cotton, 2001; Snyder and Palmer, 2004; Pradhan et al., 2007; Navigli et al., 2013; Moro and Navigli, 2015), standardized to the same format and sense inventory (i.e. WordNet 3.0).

Typically, there are two kinds of approach for WSD: supervised (which make use of sense-annotated training data) and knowledge-based (which make use of the properties of lexical resources).

Supervised: The most widely used training corpus used is SemCor, with 226,036 sense annotations from 352 documents manually annotated. All supervised systems in the evaluation table are trained on SemCor. Some supervised methods, particularly neural architectures, usually employ the SemEval 2007 dataset as development set (marked by *). The most usual baseline is the Most Frequent Sense (MFS) heuristic, which selects for each target word the most frequent sense in the training data.

Knowledge-based: Knowledge-based systems usually exploit WordNet or BabelNet as semantic network. The first sense given by the underlying sense inventory (i.e. WordNet 3.0) is included as a baseline.

The main evaluation measure is F1-score.

Example Task 10 Other Tasks (Kaggle datasets, NLP)

- https://www.kaggle.com/datasets
- https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge

Toxic Comment Classification Challenge

Identify and classify toxic online comments

Overview Data Code Models Discussion Leaderboard Rules

Overview

Start
Dec 19, 2017

Close
Mar 21, 2018

Merger & Entry

Description

Discussing things you care about can be difficult. The threat of abuse and harassment online means that many people stop expressing themselves and give up on seeking different opinions. Platforms struggle to effectively facilitate conversations, leading many communities to limit or completely shut down user comments.

The Conversation AI team, a research initiative founded by Jigsaw and Google (both a part of Alphabet) are working on tools to help improve online conversation. One area of focus is the study of negative online behaviors, like toxic comments (i.e. comments that are rude, disrespectful or otherwise likely to make someone leave a discussion). So far they've built a range of publicly available models served through the Perspective API, including toxicity. But the current models still make errors, and they don't allow users to select which types of toxicity they're interested in finding (e.g. some platforms may be fine with profanity, but not with other types of toxic content).

In this competition, you're challenged to build a multi-headed model that's capable of detecting different types of of toxicity like threats, obscenity, insults, and identity-based hate better than Perspective's current models. You'll be using a dataset of comments from Wikipedia's talk page edits. Improvements to the current model will hopefully help online discussion become more productive and respectful.

Datasets

+ New Dataset

Search 3,458 datasets

Filters

NLP ×

3,458 Datasets

Most Votes

COVID-19 Open Research Dataset Challenge (CORD-19)
Allen Institute For AI · Updated 2 years ago
Usability 8.8 · 717120 Files (JSON, CSV, other) · 20 GB
10778 Gold

Brazilian E-Commerce Public Dataset by Olist
Olist · Updated 3 years ago
Usability **10.0** · 9 Files (CSV) · 45 MB
3092 Gold

Women's E-Commerce Clothing Reviews
nicapotato · Updated 6 years ago
Usability 8.8 · 1 File (CSV) · 3 MB
1043 Gold

News Headlines Dataset For Sarcasm Detection
Rishabh Misra · Updated 5 years ago
Usability **10.0** · 2 Files (JSON) · 3 MB
968 Gold

Amazon Reviews for Sentiment Analysis
Adam Bittlingmayer · Updated 4 years ago
Usability 6.9 · 2 Files (other) · 517 MB
925 Gold

News Category Dataset
Rishabh Misra · Updated 2 years ago
Usability **10.0** · 1 File (JSON) · 28 MB
730 Gold

Coronavirus tweets NLP - Text Classification
Aman Miglani · Updated 4 years ago
Usability **10.0** · 2 Files (CSV) · 5 MB
656 Gold